Application Deployment with Terraform and CI/CD

Objective

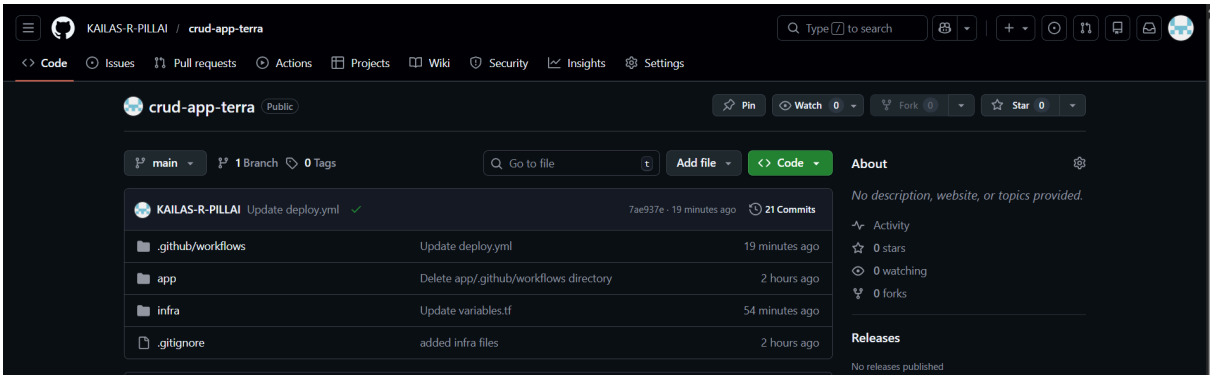
- Set up a basic infrastructure where pushing code automatically.
- Deploys infrastructure (VPC, ECS, ALB, IAM roles, etc.) using Terraform.
- Builds and deploys a containerized app to AWS ECS using CI/CD.

Repository Structure

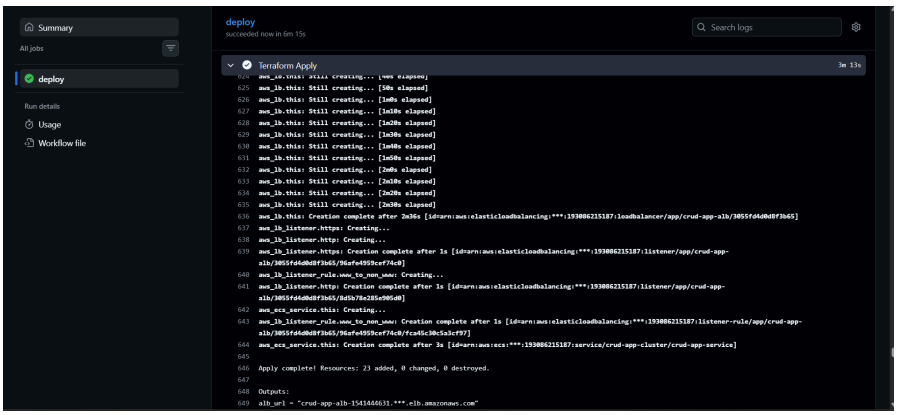
- app → Application source code
- infra → Terraform code for AWS resources
- .github/ workflows/ deploy.yml → GitHub Actions CI/CD workflow

Setup Process

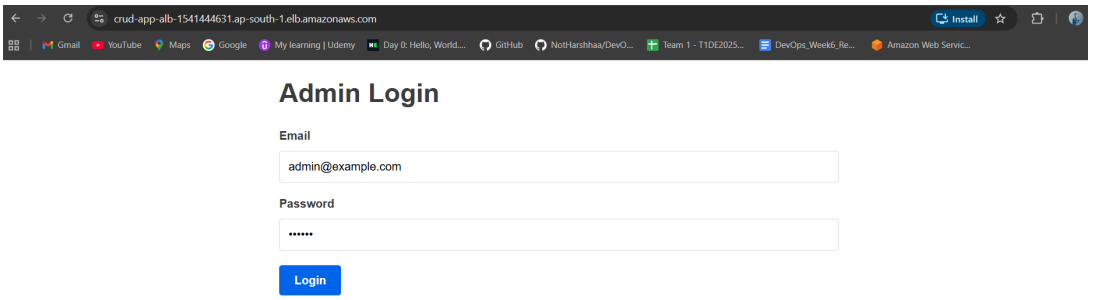
- Created terraform modules for inside a folder called infra:
  - VPC
  - IAM
  - Security Groups
  - ALB
  - ECS
  - Cloudwatch Log Groups
- Defined the ECR repository latest image URI and the certificate URI in the variables.tf
- Created the Dockerfile for the app in the app folder and pushed it to the ECR repository named crud-app with the latest tag.
- Pushed the whole code to github. Created deploy.yml in the .github/workflows which will automatically build docker image from the app directory and tag them to latest and automate terraform init, plan and apply.



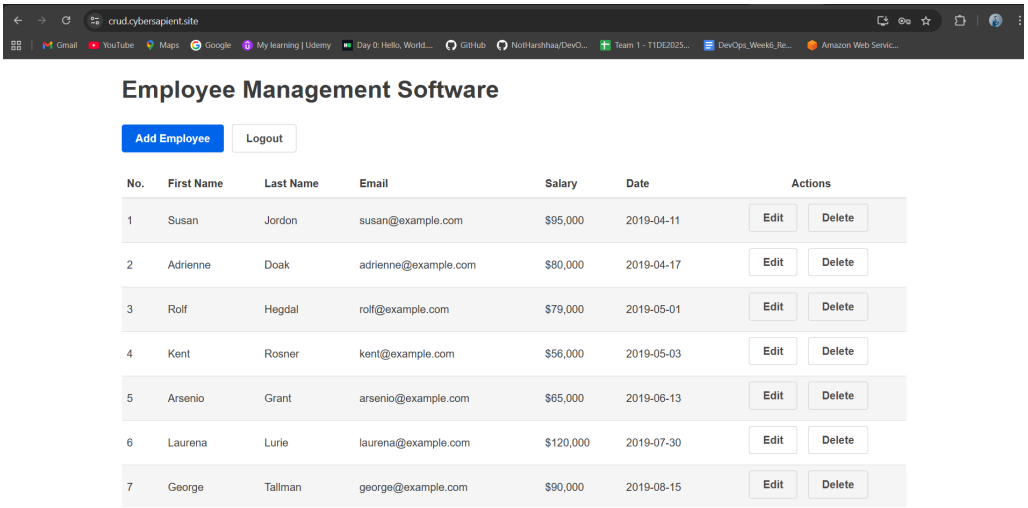
- After the deploy.yml stage was successful. Verified that every module was successfully provisioned.



- Verified that the application was accessible via ALB DNS Name and found to be successfully available.



- At last created records in the hosted zone and set alias to classic load balancer to make the application accessible via www.crud.cybersapient.site



- **Deliverables**
  - GitHub repository with code and workflows:
    - **https://github.com/KAILAS-R-PILLAI/crud-app-terra**
  - Deployed application URL:
    - **www.crud.cybersapient.site**